

YUVA INTERNSHIP PROJECT

DATA CLEANING AND ANALYSIS USING R

Submitted by : GUNAL.K

Internship : Yuva internship

Software used : R and RStudio

Project type : Data cleaning and analysis

INTRODUCTION

This project focuses on data cleaning and basic data analysis using R programming. The dataset was analysed using RStudio to understand the data and obtained useful statistical results.

In this project, R programming and RStudio were used to clean and analyse the given dataset. basic statistical function were used to understand the data and obtain useful result

The project also help in understanding the structure of the dataset, the number of records and variables, and important values such as average age and average fare.

Objectives

The main objective of this project are :

- To understand the given dataset and its structure.
- To clean and organize the data for analysis.
- To use R programing for basic data analysis.
- To calculate the average age and average fare.
- To identify the number of records and variables in the dataset.
- To understand and interpret the result obtained from the analysis

Tools used

1) R programing

Used for data cleaning, processing, and analysis.

2) RStudio

Used to write, run, and manage the R code

3) Dataset

The given dataset was used for performing the analysis

Dataset Description

The dataset contains 891 records and 12 variables.

It was examined using RStudio to understand its structure and perform basic analysis. The dataset was used to calculate values such as the average age and average fare.

Dataset Details:

- ▶ Number of records: 891
- ▶ Number of variables: 12
- ▶ Average age: 29.36
- ▶ Average fare: 32.20

Methodology

The project was completed using the following steps:

- 1) The dataset was loaded into RStudio.
- 2) The dataset was examined to understand its structure.
- 3) The data was cleaned and prepared for analysis.
- 4) Basic statistical calculations were performed using R.
- 5) The number of records and variables was identified.
- 6) The results were obtained and interpreted.

R code

The following R code was used to perform the data cleaning and basic analysis of the dataset.

```
RGui - [R Console]
File Edit View Misc Packages Windows Help

R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

>
> R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Error: unexpected symbol in "R version"
> Copyright (C) 2026 The R Foundation for Statistical Computing
Error: unexpected numeric constant in "Copyright (C) 2026"
> Platform: x86_64-w64-mingw32/x64
Error: object 'Platform' not found
>
> R is free software and comes with ABSOLUTELY NO WARRANTY.
Error: unexpected symbol in "R is"
> You are welcome to redistribute it under certain conditions.
Error: unexpected symbol in "You are"
> Type 'license()' or 'licence()' for distribution details.
Error: unexpected string constant in "Type 'license()'"
>
> Natural language support but running in an English locale
Error: unexpected symbol in " Natural language"
>
> R is a collaborative project with many contributors.
Error: unexpected symbol in "R is"
> Type 'contributors()' for more information and
Error: unexpected string constant in "Type 'contributors()'"
> 'citation()' on how to cite R or R packages in publications.
```

```
RGui - [R Console]
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Error: unexpected symbol in "'citation()' on"
>
> Type 'demo()' for some demos, 'help()' for on-line help, or
Error: unexpected string constant in "Type 'demo()'"
> 'help.start()' for an HTML browser interface to help.
Error: unexpected 'for' in "'help.start()' for"
> Type 'q()' to quit R.
Error: unexpected string constant in "Type 'q()'"
>
> > data<-read.csv(file.choose())
Error: unexpected '>' in ">"
> > head(data)
Error: unexpected '>' in ">"
> PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
Error: unexpected symbol in " PassengerId Survived"
> 1 0 3 Braund, Mr. Owen Harris male 22 1 0 A/5 21171 7.2500 S
Error: unexpected numeric constant in "1 1"
> 2 1 3 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female 38 1 0 PC 17599 71.2833 C85 C
Error: unexpected numeric constant in "2 2"
> 3 1 3 Heikkinen, Miss. Laina female 26 0 0 STON/O2. 3101282 7.9250 S
Error: unexpected numeric constant in "3 3"
> 4 1 3 Futrelle, Mrs. Jacques Heath (Lily May Peel) female 35 1 0 113803 53.1000 C123 S
Error: unexpected numeric constant in "4 4"
> 5 0 3 Allen, Mr. William Henry male 35 0 0 373450 8.0500 S
Error: unexpected numeric constant in "5 5"
> 6 0 3 Moran, Mr. James male NA 0 0 330877 8.4583 Q
Error: unexpected numeric constant in "6 6"
> > str(data)
Error: unexpected '>' in ">"
> 'data.frame': 891 obs. of 12 variables:
Error: unexpected symbol in "'data.frame': 891 obs."
> $ PassengerId: int 1 2 3 4 5 6 7 8 9 10 ...
Error: unexpected '$' in "$"
> $ Survived : int 0 1 1 1 0 0 0 1 1 ...
Error: unexpected '$' in "$"
> $ Pclass : int 3 1 3 1 3 3 1 3 3 2 ...
Error: unexpected '$' in "$"
> $ Name : chr "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)" "Heikkinen, Miss. Laina" "Futrelle, Mrs. Jacques Heath (Lily May Peel)" ...
Error: unexpected '$' in "$"
> $ Sex : chr "male" "female" "female" "female" ...
Error: unexpected '$' in "$"
> $ Age : num 22 38 26 35 35 NA 54 2 27 14 ...
Error: unexpected '$' in "$"
> $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...
```

```
RGU - [R Console]
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Error: unexpected '$' in "$"
> $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...
Error: unexpected '$' in "$"
> $ Parch : int 0 0 0 0 0 0 0 1 2 0 ...
Error: unexpected '$' in "$"
> $ Ticket : chr "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
Error: unexpected '$' in "$"
> $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
Error: unexpected '$' in "$"
> $ Cabin : chr "" "C85" "" "C123" ...
Error: unexpected '$' in "$"
> $ Embarked : chr "S" "C" "S" "S" ...
Error: unexpected '$' in "$"
> > colSums(is.na(data))
Error: unexpected '>' in ">"
> PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
Error: unexpected symbol in "PassengerId Survived"
> 0 0 0 0 0 177 0 0 0 0 0 0
Error: unexpected numeric constant in "0 0"
> > data$Age[is.na(data$Age)]<-median(data$Age.narm = true)
Error: unexpected '>' in ">"
> Error: unexpected '=' in "data$Age[is.na(data$Age)]<-median(data$Age.narm ="
Error: unexpected string constant in "Error: unexpected '='"
> > data$Age[is.na(data$Age)]<-median(data$Age.narm=TRUE)
Error: unexpected '>' in ">"
> Error: unexpected '=' in "data$Age[is.na(data$Age)]<-median(data$Age.narm="
Error: unexpected string constant in "Error: unexpected '='"
> > data$Age[is.na(data$Age)]<-median(data$Age.narm=TRUE)
Error: unexpected '>' in ">"
> > colSums(is.na(data))
Error: unexpected '>' in ">"
> PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
Error: unexpected symbol in "PassengerId Survived"
> 0 0 0 0 0 0 0 0 0 0 0 0
Error: unexpected numeric constant in "0 0"
> > summary(data)
Error: unexpected '>' in ">"
> PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin
Error: unexpected symbol in "PassengerId Survived"
> Min. : 1.0 Min. :0.0000 Min. :1.000 Length :891 Length :891 Min. : 0.42 Min. :0.000 Min. :0.0000 Length :891 Min. : 0.00 Length :891
```

```
RGU - [R Console]
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Error: unexpected symbol in "Min. : 1.0 Min."
> 1st Qu.:223.5 1st Qu.:0.0000 1st Qu.:2.000 N.unique :891 N.unique : 2 1st Qu.:22.00 1st Qu.:0.000 1st Qu.:0.0000 N.unique :681 1st Qu.: 7.91 N.unique :148
Error: unexpected symbol in "1st"
> Median :446.0 Median :0.0000 Median :3.000 N.blank : 0 N.blank : 0 Median :28.00 Median :0.000 Median :0.0000 N.blank : 0 Median :14.45 N.blank :687
Error: unexpected symbol in "Median :446.0 Median"
> Mean :446.0 Mean :0.3838 Mean :2.309 Min.nchar: 12 Min.nchar: 4 Mean :29.36 Mean :0.523 Mean :0.3816 Min.nchar: 3 Mean :32.20 Min.nchar: 0
Error: unexpected symbol in "Mean :446.0 Mean"
> 3rd Qu.:1668.5 3rd Qu.:1.0000 3rd Qu.:3.000 Max.nchar: 82 Max.nchar: 6 3rd Qu.:35.00 3rd Qu.:1.000 3rd Qu.:0.0000 Max.nchar: 18 3rd Qu.: 31.00 Max.nchar: 15
Error: unexpected symbol in "3rd"
> Max. :891.0 Max. :1.0000 Max. :3.000
Error: unexpected symbol in "Max. :891.0 Max."
> Embarked
Error: object 'Embarked' not found
> Length :891
Error: object 'Length' not found
> N.unique : 4
Error: object 'N.unique' not found
> N.blank : 2
Error: object 'N.blank' not found
> Min.nchar: 0
Error: object 'Min.nchar' not found
> Max.nchar: 1
Error: object 'Max.nchar' not found
>
> > mean(data$Age)
Error: unexpected '>' in ">"
> [1] 29.36158
Error: unexpected '[' in "["
> > mean(data$Fare)
Error: unexpected '>' in ">"
> [1] 32.20421
Error: unexpected '[' in "["
> > nrow(data)
Error: unexpected '>' in ">"
> [1] 891
Error: unexpected '[' in "["
> > ncol(data)
Error: unexpected '>' in ">"
> [1] 12
Error: unexpected '[' in "["
> > write.csv(data,"cleaned_train.csv",row.names=FALSE)
Error: unexpected '>' in ">"
>
Error: unexpected '>' in ">"
```

Output and result

The R programing was executed successfully and the following result were obtained:

- ▶ Average Age: 29.36
- ▶ Average Fare: 32.20
- ▶ Number of records: 891
- ▶ Number of variable: 12

The result shows the basic statical information and structure of the dataset

```
RGU - [R Console]
File Edit View Misc Packages Windows Help

> 1st Qu.:1223.5 1st Qu.:0.0000 1st Qu.:2.000 N.unique :891 N.unique : 2 1st Qu.:22.00 1st Qu.:0.000 1st Qu.:0.0000 N.unique :681 1st Qu.: 7.91 N.unique :148
Error: unexpected symbol in " 1st"
> Median :446.0 Median :0.0000 Median :3.000 N.blank : 0 N.blank : 0 Median :28.00 Median :0.000 Median :0.0000 N.blank : 0 Median : 14.45 N.blank :687
Error: unexpected symbol in " Median"
> Mean :446.0 Mean :0.3838 Mean :2.309 Min.nchar: 12 Min.nchar: 4 Mean :29.36 Mean :0.523 Mean :0.3816 Min.nchar: 3 Mean : 32.20 Min.nchar: 0
Error: unexpected symbol in " Mean"
> 3rd Qu.:668.5 3rd Qu.:1.0000 3rd Qu.:3.000 Max.nchar: 82 Max.nchar: 6 3rd Qu.:35.00 3rd Qu.:1.000 3rd Qu.:0.0000 Max.nchar: 18 3rd Qu.: 31.00 Max.nchar: 15
Error: unexpected symbol in " 3rd"
> Max. :891.0 Max. :1.0000 Max. :3.000 Max. :80.00 Max. :8.000 Max. :6.0000 Max. :512.33
Error: unexpected symbol in " Max."
> Embarked
Error: object 'Embarked' not found
> Length :891
Error: object 'Length' not found
> N.unique : 4
Error: object 'N.unique' not found
> N.blank : 2
Error: object 'N.blank' not found
> Min.nchar: 0
Error: object 'Min.nchar' not found
> Max.nchar: 1
Error: object 'Max.nchar' not found
>
> > mean(data$Age)
Error: unexpected '>' in ">"
> [1] 29.36158
Error: unexpected '[' in "["
> > mean(data$Fare)
Error: unexpected '>' in ">"
> [1] 32.20421
Error: unexpected '[' in "["
> > nrow(data)
Error: unexpected '>' in ">"
> [1] 891
Error: unexpected '[' in "["
> > ncol(data)
Error: unexpected '>' in ">"
> [1] 12
Error: unexpected '[' in "["
> > write.csv(data,"cleaned_train.csv",row.names=FALSE)
Error: unexpected '>' in ">"
>
Error: unexpected '>' in ">"
>
```

Conclusion

This project helped in understanding data cleaning and basic data analysis using R and RStudio. The dataset was successfully analysed, and important statistical results were obtained. The project also improved understanding of how R can be used to work with real – world data.